

Lightweight Post-processing of Multi-model Subseasonal Rainfall Forecasts over India: A Leakage-Safe 2025 Benchmark

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Abstract

Reliable weekly rainfall guidance at subseasonal leads can support climate adaptation decisions, yet heterogeneous model calendars, grids, climatologies, and test reuse make regional comparisons fragile. We construct a valid-date- aligned benchmark of seven physics, data-driven, and hybrid systems over India for JJAS initializations and lead Weeks 1–6. All forecasts and India Meteorological Department (IMD) observations use one 1.5-degree grid, one 1991–2019 IMD climatology, and identical fractional-area support. We freeze training (2020–2023), validation (2024), metrics, hyperparameters, and ablations before opening 35 common 2025 initializations. An XGBoost adaptation of PiggyCast raises mean case-level spatial ACC from 0.307 for equal weighting to 0.347 (paired difference 0.040, 95% moving-block interval [0.011, 0.070]) and reduces RMSE from 5.174 to 4.989 mm day⁻¹. Forecast-conditioned features substantially outperform a location/calendar-only ablation. However, the ACC difference from the validation-selected individual system is 0.052 [-0.002, 0.093], failing our predeclared headline gate; ACC also deteriorates relative to equal weighting at Weeks 5–6 and in east/northeast India. Lightweight mixing is therefore promising but not uniformly superior. The case-level metrics, uncertainty intervals, frozen protocol, trained models, and hashes make this bounded result fully auditable.

1 Introduction

The two-to-six-week range bridges weather prediction and seasonal outlooks, with potential value for water management, agriculture, heat and flood preparedness, and other adaptation decisions. The WWRP/WCRP S2S database made multi-centre forecast comparison possible [Vitart et al., 2017], while recent AI systems such as FuXi-S2S [Chen et al., 2024], NeuralGCM [Kochkov et al., 2024], and DLESyM [Cresswell-Clay et al., 2024] expand the set of forecast architectures. Diversity alone does not guarantee that a multi-model mean is optimal, particularly for monsoon rainfall and spatially heterogeneous model biases.

We ask a deliberately narrow question: can data-efficient post-processing improve weekly India rainfall forecasts without retraining the underlying systems? Our contributions are: (i) a seven-system benchmark with exact initialization and valid-date alignment; (ii) a regional adaptation of PiggyCast [Kimani et al., 2025] using forecast-conditioned gradient-boosted trees; and (iii) a confirmatory 2025 evaluation with predeclared claim

Table 1: Forecast inventory. The 2024 common validation intersection contains 30 cases; 2020–2023 and 2025 contain 35 per year.

System	Type	Members	2020–23 cases/year	2025 cases
CMA operational	physics	4	35	35
DLESyM v0	AI Earth system	1	35	35
ECMWF operational	physics	51 or 101	35	35
FuXi-S2S	AI subseasonal	50	35	35
NCEP operational	physics	16	35	35
NeuralGCM	hybrid AI-physics	10	35	35
UKMO operational	physics	4	35	35

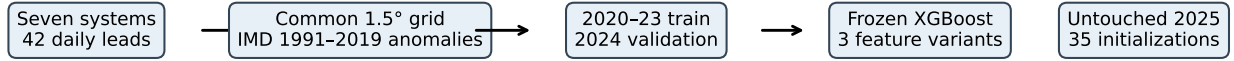
gates and paired uncertainty. The outcome is useful precisely because it is not uniformly positive: adaptive mixing clears the equal-weight and feature-ablation gates but not the stronger validation-selected-model gate.

2 Data and harmonization

Forecasts. We evaluate CMA, DLESyM v0, ECMWF, FuXi-S2S, NCEP, NeuralGCM, and UKMO weekly ensemble-mean precipitation. Table 1 summarizes the products. Each case has 42 complete daily rates. Lead Week 1 averages intervals ending initialization+1 through +7 days; Week 6 averages +36 through +42. We retain small negative finite precipitation increments produced by differencing rather than silently clipping them. Five systems use immutable standardized 2025 stores; DLESyM and NeuralGCM use native 2025 files whose units, coordinates, and period endpoints match the same archive contract.

Truth, grid, and anomalies. IMD gauge-based daily rainfall [Pai et al., 2014] is conservatively aggregated to a 27×27, 1.5-degree grid spanning 0–39°N and 60–99°E. Grid-cell weights multiply spherical area, fractional India or regional overlap, and IMD observation fraction. A fixed 31-day-window IMD 1991–2019 normal is averaged over each forecast’s exact seven valid dates and subtracted from both forecast and observation. No model-specific climatology enters any ACC calculation.

Splits and coverage. We train with 140 JJAS cases from 2020–2023, use the 30-case common 2024 intersection for all selection, and test once on 35 common 2025 cases. The five missing 2024 initializations are June 6, 10, 13, 17 and August 5. Later lead windows may verify after September because season membership is defined by the initialization date.



All forecasts and observations share init+1...+42 valid-day support; test fitting is forbidden

Figure 1: Leakage-safe workflow. The 2025 forecast values were opened only after the complete protocol and metadata-only coverage preflight were frozen.

3 Methods

Baselines. We retain all seven individual ensemble means. The equal-weight baseline is the arithmetic multi-model mean. A validation-weighted mean uses non-negative, per-week inverse 2024 RMSE weights. A validation-selected baseline takes the lowest-2024-RMSE individual model separately at each lead: FuXi-S2S at Week 1 and ECMWF at Weeks 2–6.

Adaptive mixing. Following PiggyCast, we fit one XGBoost regressor [Chen and Guestrin, 2016] per lead week. Forecast-only inputs comprise seven forecast anomalies and six real ensemble-spread fields (DLESyM is deterministic). The full variant adds latitude, longitude, and sine/cosine of verification-midpoint day of year. A location/calendar-only ablation removes all forecast information. Hyperparameters are fixed at maximum depth 4, learning rate 0.03, subsample 0.8, column subsample 0.9, and at most 600 trees. Early stopping on 2024 chooses the tree count; a fresh estimator with that count is refit on 2020–2024, then applied to 2025. Area weights are used during fitting and validation.

Verification and uncertainty. The primary score is case-level area-weighted spatial anomaly correlation (ACC), averaged equally across initialization-week cases. Secondary metrics are RMSE, MAE, signed bias, and wet-area fraction error at 1 mm day^{-1} . We report paired percentile intervals from 2,000 circular moving-block draws over ordered initialization dates, retaining all six leads together. The primary block length is four initializations; lengths 2 and 8 are sensitivities. Intervals are descriptive effect intervals rather than p -values.

4 Confirmatory 2025 results

4.1 Overall and lead-dependent skill

Full PiggyCast attains ACC 0.347, RMSE 4.989, MAE 3.603, and bias $-0.303 \text{ mm day}^{-1}$, compared with equal-weight ACC 0.307, RMSE 5.174, MAE 3.716, and bias -0.428 . The paired ACC gain over equal weight is 0.040 [0.011, 0.070], and its RMSE difference is -0.186 [-0.290 , -0.096] mm day^{-1} . Inverse-validation-RMSE weighting yields a smaller ACC improvement, 0.313 versus 0.307.

Figure 2 shows strong lead dependence. Full PiggyCast improves ACC over equal weighting at Weeks 1–4 (by 0.057, 0.064, 0.070, and 0.099), is nearly tied at Week 5 (-0.001), and is worse at Week 6 (-0.047). Its RMSE is lower at every lead. This divergence reinforces the need to report both pattern and amplitude metrics.

4.2 Gates, ablations, and regional behavior

The forecast-only variant exceeds the location/calendar-only ablation by 0.238 ACC [0.209, 0.266], showing that the gain cannot be attributed to a learned seasonal spatial normal alone. Adding calendar/location features to the forecast-only model raises overall ACC from 0.339 to 0.347 and reduces RMSE from 5.028 to 4.989.

The stronger comparison is inconclusive. Full PiggyCast exceeds the validation-selected model in mean ACC by 0.052, but its interval [-0.002 , 0.093] overlaps zero. Our frozen gate therefore prohibits the headline claim that adaptive mixing is robustly better than the selected individual system, despite favorable RMSE and bias guards.

Regional mean ACC differences relative to equal weighting are $+0.128$ in northwest India, $+0.060$ in the south peninsula, $+0.011$ in central India, and -0.097 in east/northeast India. Figure 3 displays paired overall and regional intervals. The three-of-four positive-region gate passes, but the east/northeast failure precludes language implying geographically uniform benefit.

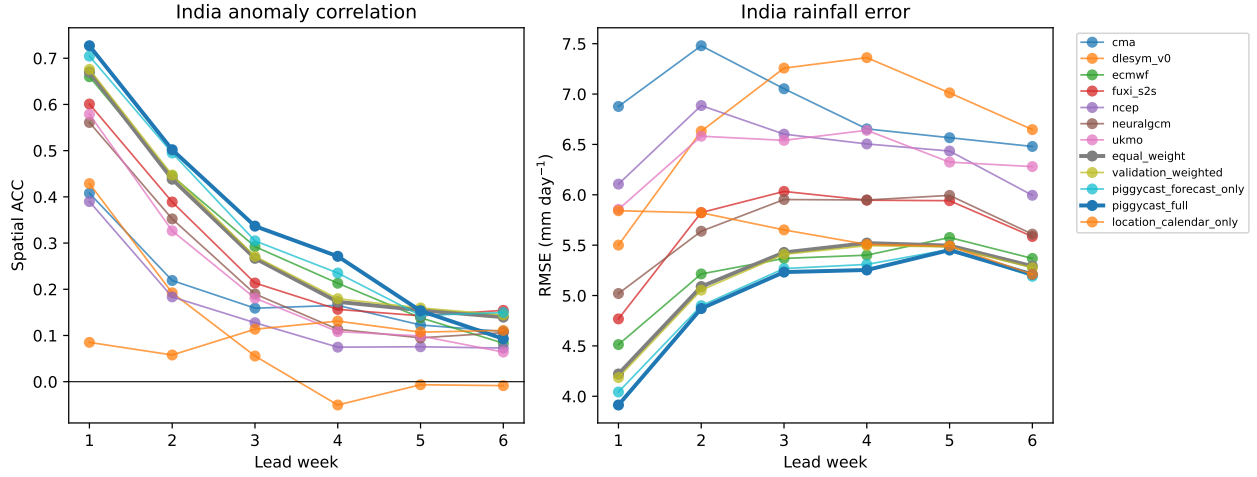


Figure 2: India-area-weighted case-mean spatial ACC and RMSE by lead.

Table 2: Overall 2025 scores across 35 initializations and six lead weeks. Wet-area error is forecast minus observed weighted wet fraction.

Method	ACC $\uparrow$	RMSE $\downarrow$	MAE $\downarrow$	Bias	Wet-area error
piggycast_full	0.347	4.989	3.603	-0.303	0.113
piggycast_forecast_only	0.339	5.028	3.626	-0.288	0.117
validation_weighted	0.313	5.150	3.700	-0.397	0.114
equal_weight	0.307	5.174	3.715	-0.428	0.114
ecmwf	0.305	5.240	3.822	0.253	0.107
validation_selected_model	0.296	5.283	3.845	0.157	0.097
fuxi_s2s	0.276	5.683	4.014	-0.363	0.047
neuralgcm	0.236	5.694	4.094	-0.429	0.099
ukmo	0.226	6.370	4.522	0.328	0.044
cma	0.197	6.852	4.839	-0.364	0.052
ncep	0.154	6.421	4.538	-1.471	-0.013
dlesym_v0	0.102	6.735	4.815	-0.950	0.010
location_calendar_only	0.101	5.587	4.091	-0.402	0.144

Table 3: Predeclared paired ACC comparisons with 95% moving-block intervals.

Paired ACC comparison	Mean	2.5th pct.	97.5th pct.
Full – equal weight	0.040	0.011	0.070
Full – validation-selected	0.052	-0.002	0.093
Forecast-only – location/calendar	0.238	0.209	0.266

5 Discussion

The results support a bounded operational proposition: a compact post-processor can extract complementary information from heterogeneous forecast systems and outperform an uncalibrated equal-weight mean without changing the underlying models. The forecast-only ablation provides direct evidence that the method uses forecast state, while the lower RMSE and less negative bias argue against an ACC-only artifact.

The failed selected-model gate is equally important. First, 35 test initializations give limited precision under serial dependence. The comparison changes with bootstrap block length: its lower endpoint is +0.009 for length 2 but -0.002 at lengths 4 and 8, whereas the equal-weight comparison remains positive at all three lengths. Second, selected tree counts collapse from hundreds at

Weeks 1–3 to 8–12 at Weeks 5–6, consistent with weak conditional signal at long leads. Third, geographic transfer is uneven, with a clear east/northeast loss. A production system should therefore retain per-lead and regional guardrails rather than deploying one national aggregate decision.

This study evaluates deterministic ensemble means and does not assess probabilistic calibration, extremes, onset timing, or decision value. Model versions and operational ensemble sizes differ, and the 2024 validation archive has a five-date gap. The fixed IMD normal avoids cross-method climatology leakage but a single 1991–2019 baseline cannot resolve nonstationarity. Finally, the compact FuXi neural correction planned as a complementary method is absent: its existing frozen evaluator uses initialization+0–6 day labels, inconsistent with the benchmark’s +1–7 period endpoints. We report this as a negative workflow finding rather than compare misaligned scores.

6 Conclusion

We provide a reproducible seven-system India S2S rainfall benchmark and a leakage-safe test of lightweight adaptive mixing. PiggyCast improves overall 2025 ACC

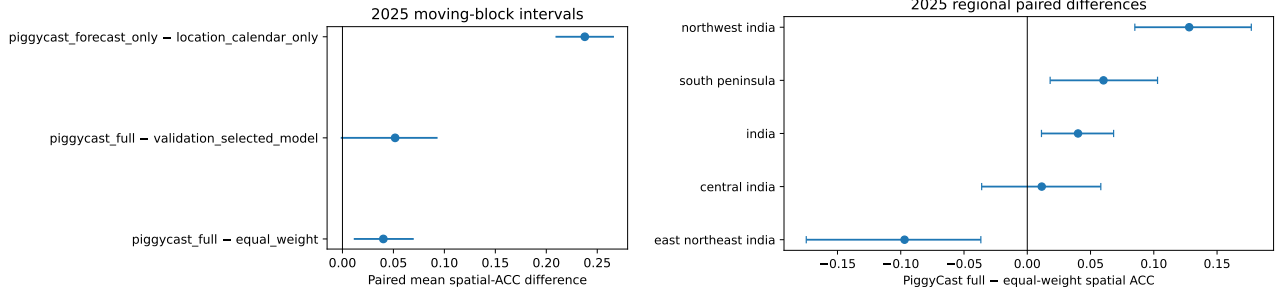


Figure 3: Paired 2025 ACC differences. Bars are 95% circular moving-block percentile intervals over initialization dates; all six leads remain grouped.

and RMSE relative to equal weighting and clearly outperforms a location/calendar-only ablation. It does not clear the predeclared uncertainty gate against the validation-selected individual system and loses ACC at the longest leads and in east/northeast India. The appropriate conclusion is not that post-processing is uniformly best, but that forecast-conditioned mixing has measurable, region- and lead-dependent value that warrants broader testing.

Reproducibility statement

The supplementary bundle contains the frozen JSON protocol, exact coverage, case-level metrics, aggregate tables, paired intervals, bootstrap sensitivity, 18 serialized XGBoost models, figures, source and artifact hashes, and an audit report. The audit independently regenerates every aggregate score from the case table. No 2025 fitting, threshold selection, feature selection, or retry selection is performed.

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A Supplementary protocol and diagnostics

A.1 Complete coverage

The common initialization counts are 35 in each of 2020–2023, 30 in 2024, and 35 in 2025. Exact ISO dates are stored in `coverage.csv`; per-model availability and source metadata hashes are stored in `preflight/coverage_preflight.csv` and `preflight/preflight.json`. The metadata-only preflight records `forecast_values_read=false`.

A.2 Claim gates

The headline requires positive lower 95% ACC endpoints against both equal weighting and the validation-selected individual system, a positive forecast-only versus location-only endpoint, at most 5% relative RMSE regression, at most 0.5 mm day^{−1} absolute-bias regression, and positive regional ACC deltas in at least three

of four IMD regions. All pass except the selected-model ACC endpoint, -0.002. The machine-readable decision is in `gate_report.json`.

A.3 Regional intervals

Table 4: Full PiggyCast minus equal-weight regional ACC.

Domain: full – equal weight	Mean	2.5th pct.	97.5th pct.
india	0.040	0.011	0.068
northwest india	0.128	0.085	0.177
central india	0.011	-0.036	0.058
south peninsula	0.060	0.018	0.103
east northeast india	-0.097	-0.175	-0.037

A.4 Hyperparameters and compute

XGBoost uses squared-error loss, histogram trees, CPU execution, maximum depth 4, minimum child weight 5, learning rate 0.03, row subsampling 0.8, column subsampling 0.9, L_2 regularization 1, and seed 42. The maximum tree count is 600 with patience 50. The exact selected count, feature names, and row counts for every lead/variant are in `fit_metadata.json`. Runtime and energy were not instrumented reliably enough for a numerical emissions claim.

A.5 Neural correction exclusion

The available FuXi neural-correction control defines Week 1 as initialization through initialization+6 days, while every benchmark forecast defines daily lead 1 as the accumulation interval ending initialization+1 day. Applying its saved model or scores here would shift labels by one day. It is excluded from the common-date result and may only be revisited after training under the frozen +1–+42 endpoint contract.

A.6 Reproduction commands

```
PYTHONPATH=src python run.py preflight
PYTHONPATH=src python run.py run
PYTHONPATH=src python run.py audit
PYTHONPATH=src python make_paper.py
```

The runner refuses an existing output directory. A repeated test must use a new path and is a distinct confirmatory run, never a replacement chosen by score. SHA256 values for every saved artifact appear in `manifest.json`.